

Features

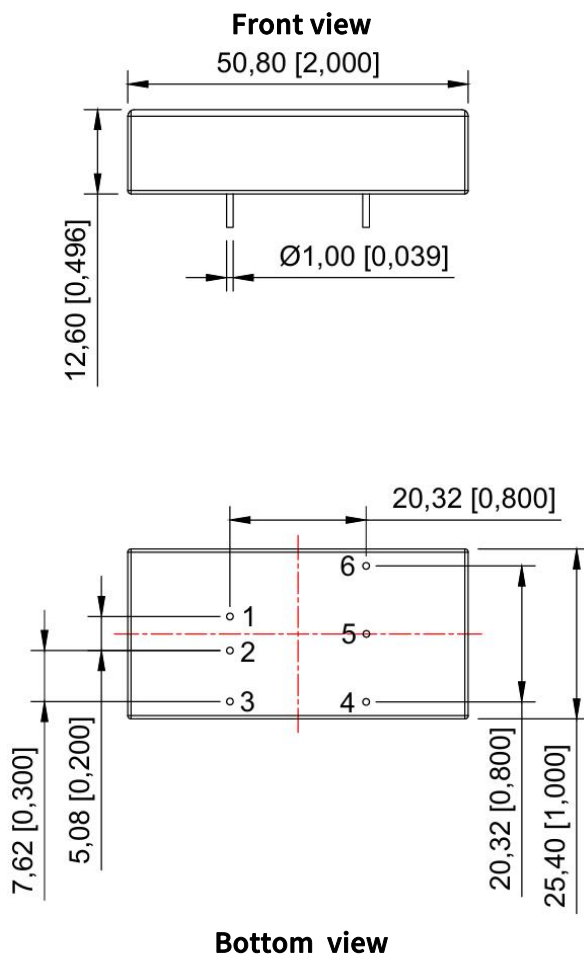
- Wide voltage input 2:1
- DIP/ radiator/Din-rail series
- Operating temperature range: $-40^{\circ}\text{C}\sim+85^{\circ}\text{C}$
- Isolation 1500VDC 0.5mA 1Minute
- Internal SMD design
- Metal shell
- Natural air cooling
- It has good shielding anti-interference performance and electromagnetic compatibility, lightning protection, output over current, short circuit protection, overheat protection, self-recovery and other functions

Product Picture



Dimensions

WRD_S_-40WH2 Series Dimensions



Note: The grid distance:2.54*2.54mm

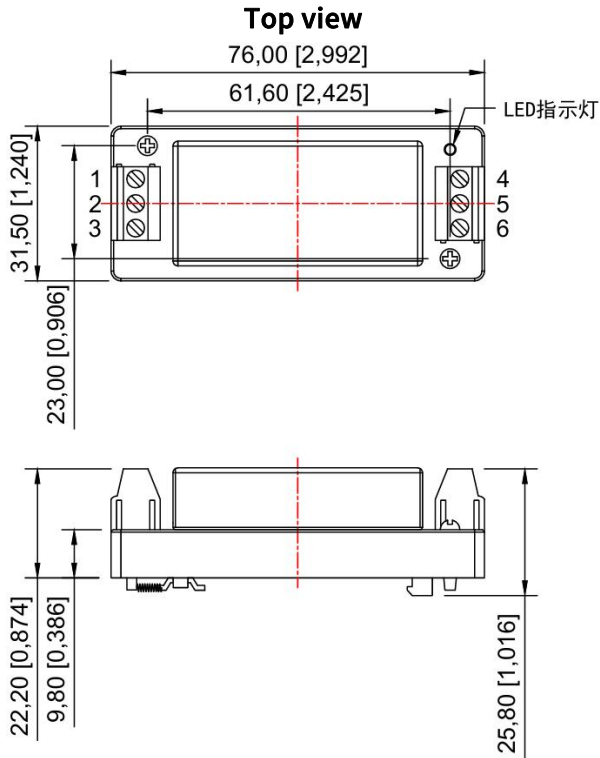
Pin mode	
Pin	Function
1	Vin
2	GND
3	CNT
4	0V
5	TRM
6	+XXVDC

Note:

Size unit: mm[inch] Pin section tolerance: $\pm 0.1[\pm 0.004]$

Unmarked tolerance: $\pm 0.25[\pm 0.01]$ The device layout is for reference only.

WRD_S_ZDK-40WH2 Series Dimensions



Front view

Pin mode	
Pin	Function
1	CNT
2	GND
3	Vin
4	0V
5	TRM
6	+XXVDC

Note:

Size unit: mm[inch]

Unmarked tolerance: $\pm 0.25[\pm 0.01]$

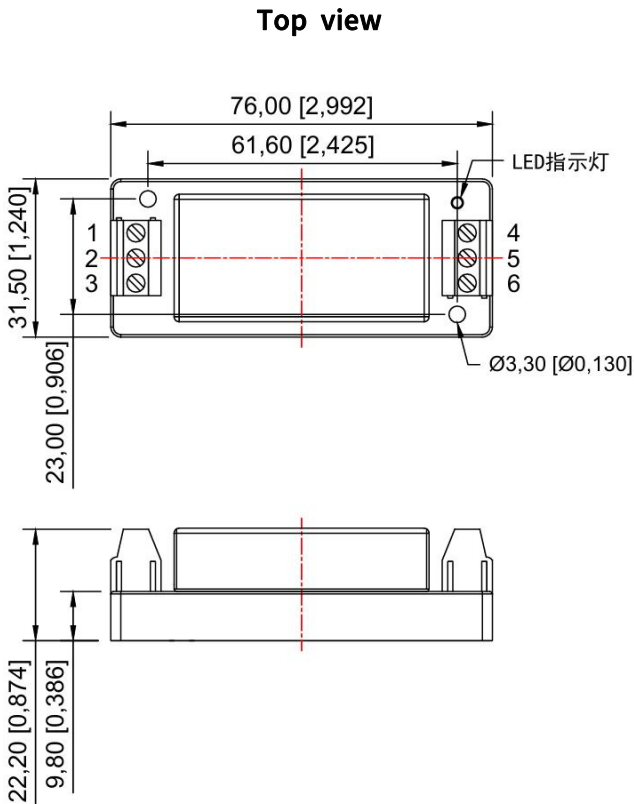
Wiring strength: 24-12 AWG

Tightening torque: Max 0.4N-m

Din-rail type: TS35

The device layout is for reference only.

WRD_S_ZD-40WH2 Series Dimensions



Front view

Pin mode	
Pin	Function
1	CNT
2	GND
3	Vin
4	0V
5	TRM
6	+XXVDC

Note:

Size unit: mm[inch]

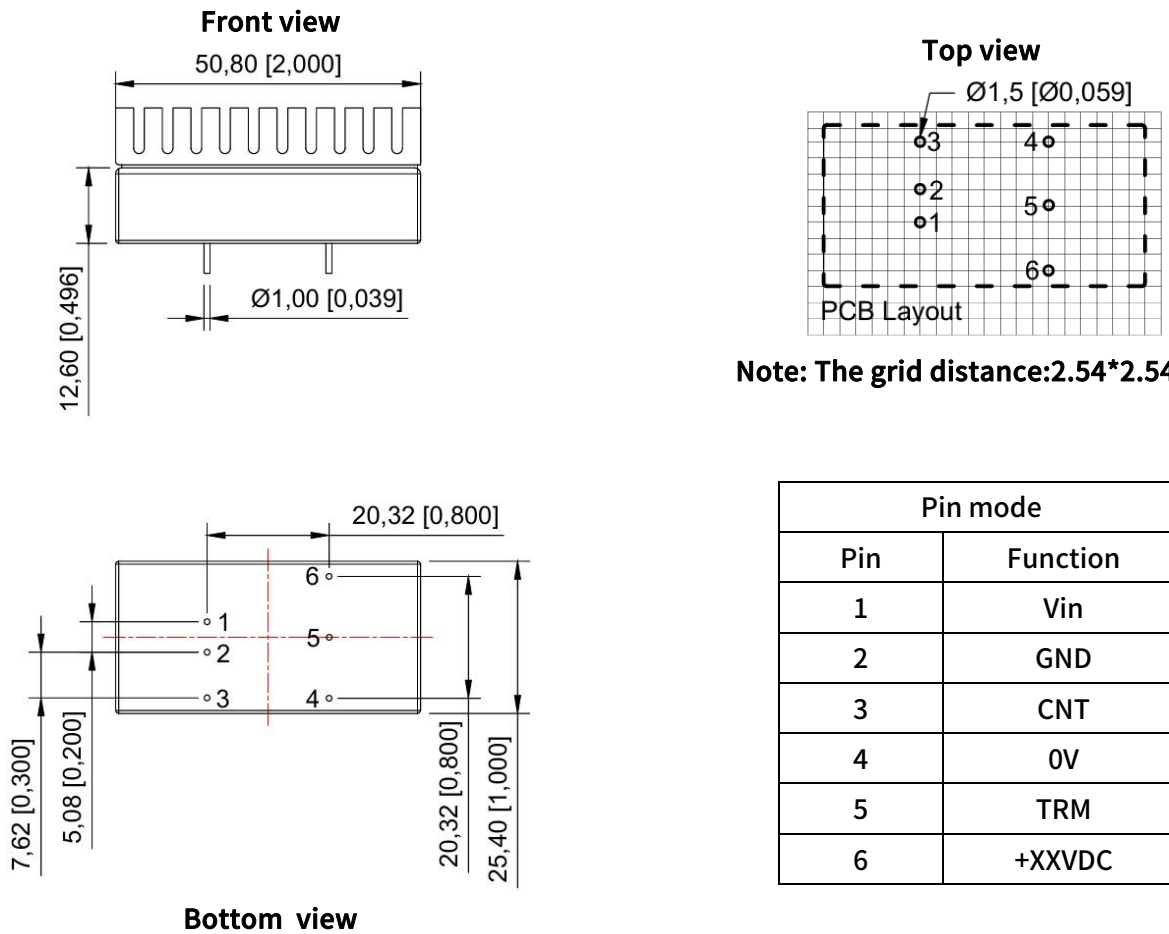
Unmarked tolerance: $\pm 0.25[\pm 0.01]$

Wiring strength: 24-12 AWG

Tightening torque: Max 0.4N-m

The device layout is for reference only.

WRD_S_-40WH2S Series Dimensions



Note:
Size unit: mm[inch]Pin section tolerance: $\pm 0.1[\pm 0.004]$
Unmarked tolerance: $\pm 0.25[\pm 0.01]$ The device layout is for reference only.

Pin mode	
Pin	Function
1	Vin
2	GND
3	CNT
4	0V
5	TRM
6	+XXVDC

Application

Railway communication, display, monitoring equipment, petrochemical, industrial control, remote DC power supply system, switching system and other communication equipment.

Selection Guide

Model	Vin (VDC)	Vout (Vo $\pm$ 2%)	Current (mA)	Efficiency (%)	Isolation (VDC)	Weight (g $\pm$ 0.5)	Certification
WRD__S3.3-40WH2(S)	12(9-18) 24(18-36) 48(36-72) 110(70-150)	3.3	12121	≥ 85	1500		
WRD__S05-40WH2(S)		5	8000	≥ 86	1500		
WRD__S12-40WH2(S)		12	3333	≥ 87	1500		
WRD__S15-40WH2(S)		15	2667	≥ 87	1500		
WRD__S18-40WH2(S)		18	2222	≥ 87	1500		
WRD__S24-40WH2(S)		24	1667	≥ 87	1500		
WRD__S28-40WH2(S)		28	1429	≥ 87	1500		
WRD__S36-40WH2(S)		36	1111	≥ 87	1500		
WRD__S48-40WH2(S)		48	833	≥ 87	1500		
WRD__S3.3ZD(K)-40WH2(S)		3.3	12121	≥ 85	1500		

WRD__S05ZD(K)-40WH2(S)		5	8000	≥ 86	1500		
WRD__S12ZD(K)-40WH2(S)		12	3333	≥ 87	1500		
WRD__S15ZD(K)-40WH2(S)		15	2667	≥ 87	1500		
WRD__S18ZD(K)-40WH2(S)		18	2222	≥ 87	1500		
WRD__S24ZD(K)-40WH2(S)		24	1667	≥ 87	1500		
WRD__S28ZD(K)-40WH2(S)		28	1429	≥ 87	1500		
WRD__S36ZD(K)-40WH2(S)		36	1111	≥ 87	1500		
WRD__S48ZD(K)-40WH2(S)		48	833	≥ 87	1500		

Note: The company for customers to customize any input and output module power supply, if you have special needs, please call our company, unless otherwise specified, input =Vi, the characteristics of the module power supply should meet the requirements of Table 1, and applicable to the full temperature range (-40°C≤Tc≤85°C)

Electrical Specifications

Specifications	Symbol	Conditions Vi , -40°C≤Tc≤85 (Unless otherwise specified)	Min	Max	Unit
Output Voltage	Vo	Full Load	Vo-2%	Vo+2%	V
Output Current	Iomax	—	—	P(Power)/ U(Output voltage)	A
Output Ripple Voltage	Vp-p	Full Load, Vi, BW=20MHz, Normal Temperature	100	150	mV
Output Noise Voltage	Vp-p	Full Load, Vi, BW=20MHz, Normal Temperature	150	250	mV
Voltage Regulation	Sv	Vimin、Vi、Vimax, Full Load	—	<2%	%
Load Adjustment	Si	Vi, Io=(10%~100%)Iomax	—	<1%	%
Efficiency	η	Vi, Full Load, Normal Temperature	85	—	%
Insulation Resistance	RI	Input-output, Insulation Voltage 500VDC	1000	—	MΩ

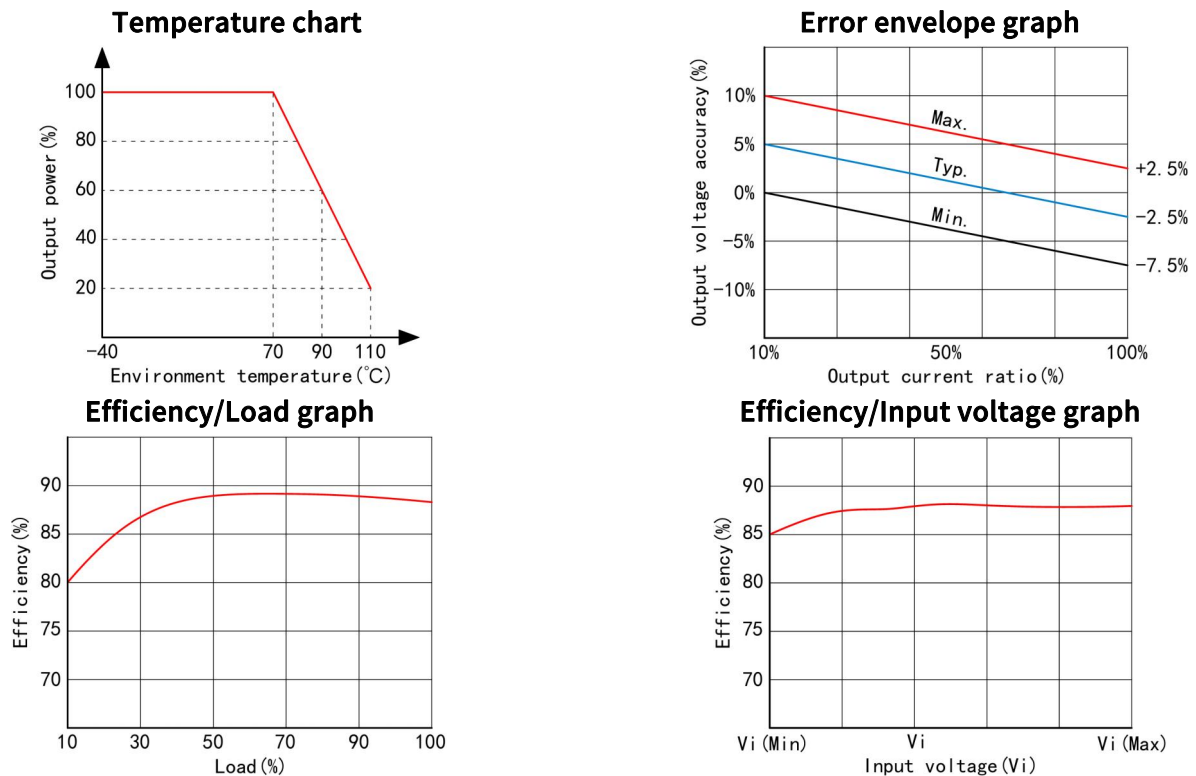
General Specifications

EMC Specifications	Magnetic Field Sensitivity Test	GB-4943
	Electrostatic Discharge Sensitivity Test	GB-4943
	Radiation Sensitivity Test	GB-4943
	Conduction Sensitivity Test	GB-4943
Temperature Drift	≤0.02%/°C	
Storage Temperature	-40°C~125°C	
Input Frequency	270KHz~400KHz	
Humidity	10%~90%RH	
Leakage Current	No	
MTBF	>500000H	

Mechanical Specifications

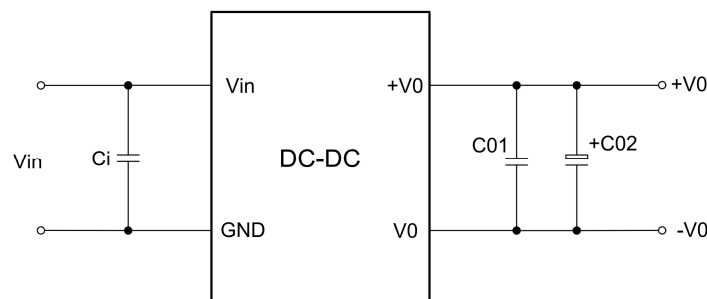
Size	50.80 x 25.40 x 12.60 mm, With radiator: 50.80 x 25.40 x 22.60 mm, ZD(K): 76.00 x 31.5 mm
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Typical Specifications Curves



Typical Application

Design Reference



Recommendation Test

Filter: In some circuits that are sensitive to noise and ripple, an external filter capacitor can be connected to the DC/DC input and output terminals to reduce the impact of ripple on the system, but the value of the filter capacitor should be appropriate, if the capacitor is too large, it is likely to cause startup problems, for each output, under the condition of ensuring safe and reliable operation, the maximum capacitance of the filter capacitor can be referred to the external capacitance table. In order to obtain very low ripple, an "LC" filter network can be connected to the input and output end of the DC/DC converter, so that the filtering effect will be better, and it should be noted that the size of the inductance value and the frequency of the "LC" filter network should be staggered from the frequency of the DC/DC module power supply to avoid mutual interference. For each output, under safe and reliable working conditions, the recommended capacitive load value is shown in (Table 1).

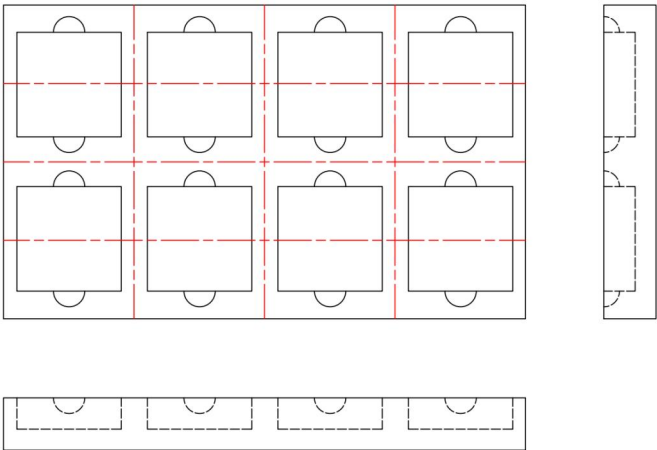
Single Input	Cout	Dual Vout	Cout
5-12VDC	22-68uF	$\pm 5\text{-}\pm 12\text{VDC}$	4.7-22uF
24-48VDC	10-47uF	$\pm 24\text{-}\pm 48\text{VDC}$	4.7-10uF

Recommended output max capacitive load value table (Table 1)

Precautions

Package

This series of modules are packed in shockproof and anti-static foam.



Transport

The package containing the module is allowed to be transported by any means of transport, which should avoid direct rain and snow and mechanical damage.

Storage

The module should be stored in a warehouse where the ambient temperature is -40 degrees ~ 125 degrees, the relative humidity is 10%~90%, and the surrounding environment is free from acidic, alkaline and other harmful gases.

Note: The above are the performance indicators of the product series listed in this manual. Some indicators of non-standard products may exceed the above requirements, so if there is any inconsistency between the manual and the product specification documents, please refer to the specification documents. If you have special needs, please contact us directly.